

Introduction to ICT

ICT (Information and Communication Technology) is the use of computers, mobile devices, the internet, and communication technologies to store, process, and share information.

Uses of ICT

- Education (online learning)
- Business (online banking, e-commerce)
- Healthcare (telemedicine)
- Communication (email, video calls)
- Entertainment

Advantages

- Fast communication
- Easy access to information
- Saves time and money
- Improves productivity

What is an Operating System (OS)?

An **Operating System (OS)** is **system software** that manages the computer's hardware and software. It acts as an **interface between the user and the computer**.

Examples: Windows, Linux, macOS, Android, iOS

Types of Operating System

1. **Single User Operating System**
 - Used by one user at a time.
 - Example: MS-DOS
2. **Multi-User Operating System**
 - Allows multiple users to use the same computer simultaneously.
 - Example: UNIX
3. **Multitasking Operating System**
 - Runs more than one program at the same time.
 - Example: Windows 10, Windows 11
4. **Mobile Operating System**
 - Designed for smartphones and tablets.
 - Examples: Android, iOS

Functions (Work) of an Operating System

- **User Interface:** Helps users interact with the computer.
- **Memory Management:** Allocates and manages computer memory.
- **File Management:** Creates, stores, copies, and deletes files.
- **Device Management:** Controls devices like printers, keyboards, and scanners.
- **Process Management:** Runs and manages programs efficiently.
- **Security:** Protects the computer using passwords and permissions.

The main desktop components of **Windows 10** are:

1. **Desktop** – The main workspace where icons, folders, and shortcuts are displayed.
2. **Start Menu** – Opens when you click the **Start** button. It provides access to apps, settings, files, and power options.
3. **Taskbar** – Located at the bottom of the screen by default. It contains:
 - Start button
 - Search box
 - Pinned applications
 - Running programs
 - Task View button
 - Notification area (system tray)
4. **Desktop Icons** – Small pictures representing files, folders, applications, or shortcuts (such as This PC and Recycle Bin).
5. **Notification Area (System Tray)** – Found at the right end of the taskbar. It displays the clock, date, network, volume, battery status (on laptops), and background app notifications.
6. **Task View Button** – Lets you view all open windows and create or switch between virtual desktops.
7. **Recycle Bin** – Stores deleted files temporarily until they are permanently deleted or restored.
8. **Background (Wallpaper)** – The image or color displayed on the desktop behind the icons.

General Maintenance Tips

- Keep the operating system and applications updated.
- Use a reliable antivirus program.
- Avoid downloading software from untrusted sources.
- Store the computer in a cool, dry, and well-ventilated place.
- Regularly back up important data to an external drive or cloud storage.
- Use a surge protector or UPS to protect against power fluctuations.

Types of Computer Viruses

1. **Boot Sector Virus**
 - Infects the computer's boot sector or master boot record.
 - Activates when the computer starts.
2. **File Infector Virus**
 - Attaches itself to executable files (such as .exe files).
 - Spreads when the infected file is opened.
3. **Macro Virus**
 - Infects documents and spreadsheets that contain macros.
 - Commonly spreads through email attachments.
4. **Multipartite Virus**
 - Infects both the boot sector and files.
 - Can spread quickly and is difficult to remove.
5. **Resident Virus**
 - Loads into the computer's memory.
 - Infects files while the operating system is running.
6. **Polymorphic Virus**
 - Changes its code each time it spreads.
 - Makes detection by antivirus software more difficult.

7. Worm

- A self-replicating program that spreads across networks without attaching to a file.

8. Trojan Horse

- Disguises itself as legitimate software.
- Can steal data or give attackers unauthorized access to the computer.

Methods to Prevent Computer Viruses

- Install and regularly update antivirus software.
- Keep the operating system and applications up to date.
- Avoid opening email attachments or links from unknown senders.
- Download software only from trusted websites and official sources.
- Scan USB drives and other external storage devices before opening files.
- Use a firewall to help block unauthorized network access.
- Avoid visiting suspicious or unsafe websites.
- Create regular backups of important files.
- Use strong passwords and enable multi-factor authentication where available.
- Do not install pirated or unlicensed software.